
The Tangent and Velocity Problems

Work these in class. Worked solutions are on the posted slide deck — each problem below names the slide to look for.

2.1 • Problem 1*slide: Sliding Q Toward P*

Fill in $m_{PQ} = \frac{x^2 - 9}{x - 3}$ for $x = 6, 4, 3.5, 3.1, 3.01$, then from the other side for $x = 0, 2, 2.5, 2.9, 2.99$.
What are the slopes doing?

2.1 • Problem 2 (worked together)*slide: Evaluating the Limit*

Why we are allowed to cancel $(x - 3)$ is the whole point, so let me do this one.